

High-Power Multi-Wavelength Hybrid Integrated Laser Array for Co-Packaged Optics

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Abstract A four-channel L-band hybrid laser array is demonstrated using an InP gain-chip array integrated on a CMOS-compatible optical interposer. Single-mode operation with >50 dB SMSR and 400 GHz spacing is achieved without active wavelength tuning. Each channel delivers over 20 dBm at 50 °C. ©2026 The Author(s)

Introduction

Silicon photonics is attracting significant interest due to its compatibility with low-cost complementary metal-oxide-semiconductor (CMOS) fabrication processes and its capability to integrate photonic and electronic components on a single chip, making it particularly attractive for hyperscale data-centre applications [1,2]. Recently, co-packaged optics (CPO) [3,4] has been widely recognized as a promising solution to address the growing bandwidth and energy-efficiency challenges in next-generation data-centre systems. However, this architecture imposes stringent requirements on optical sources, particularly in terms of high output power, scalability, and wavelength multiplexing capability. In this context, high-power multi-wavelength laser arrays are highly desirable to efficiently support parallel optical channels while minimizing system complexity. Commercial solutions typically rely on high-power distributed feedback (DFB) lasers, which have been extensively developed and offer mature performance. However, high-power DFB lasers often suffer from yield limitations due to mode instability and spatial hole burning effects [5,6]. In addition, dense wavelength-division multiplexing (DWDM) is required to expand the bandwidth of data-centre interconnects. However, achieving precise wavelength spacing in DFB arrays usually requires electron-beam lithography which increases fabrication complexity and limits scalability. Therefore, alternative laser architectures that can simultaneously deliver high power, wavelength scalability, and integration flexibility are highly required. Recently, III-V/silicon integrated lasers [7-9] have emerged as a promising solution. Compared with methods such as hetero-epitaxy or heterogeneous wafer/die bonding, flip-chip bonding method allows independent fabrication and optimization of III-V and silicon photonic components while also enabling the selection of known good dies prior to integration. Furthermore, the flexibility of photonic design on silicon platforms enables advanced functionalities in hybrid lasers, including chip-scale optical

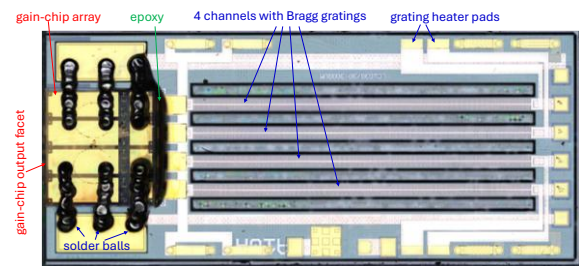


Fig. 1: Scope picture of a hybrid integrated laser array.

isolators [10,11] or even isolator-free laser operation [12,13]. Compared with single-device hybrid lasers, hybrid integrated laser arrays [14-16] further reduce the packaging cost by enabling multiple channels integration in single bonding step. This approach provides a scalable solution to realize multi-wavelength, high-power light sources within a compact footprint, which is highly desirable for CPO systems.

In this paper, we report a four-channel hybrid laser array consisting of an InP gain-chip array and a POET-patented optical interposer incorporating Bragg gratings. Single-longitudinal-mode operation is achieved for all channels with a channel spacing of 400 GHz in the L-band wavelength range. Each channel delivers more than 125 mW output power at 25 °C and approximately 100 mW at 50 °C when driven at a current of 400 mA. This represents one of the highest output powers reported for hybrid integrated lasers.

Design and Fabrication

Figure 1 illustrates the structure of a hybrid integrated laser array. The devices operate as external-cavity lasers: light generated in the gain chip is butt-coupled into the interposer waveguide and reflected by the Bragg grating, which provides wavelength-selective feedback. The reflected light is coupled back into the gain chip to form the lasing cavity, while the remaining light is emitted from the gain-chip output facet. The use of Bragg grating-based linear reflectors enables a compact and scalable layout with tight channel pitch, supporting a higher density of light sources on

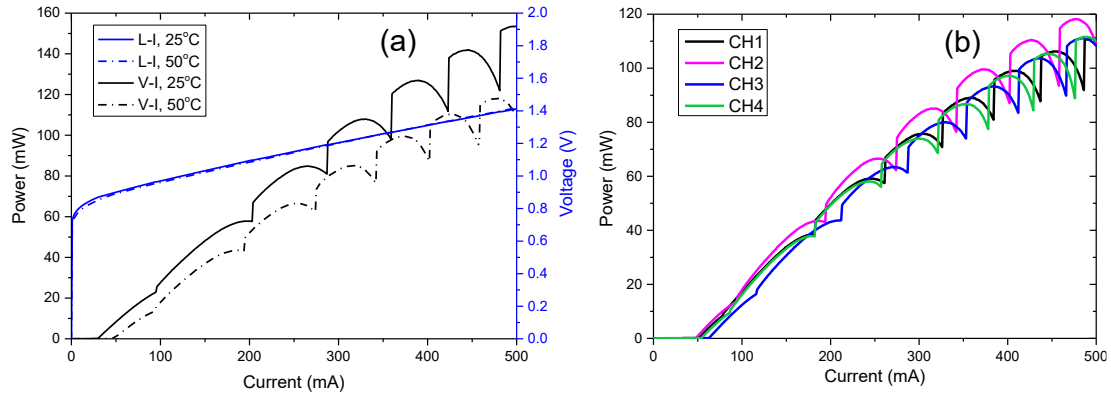


Fig. 2: (a) L-I-V curves of CH2 laser in CW mode at 25 °C and 50 °C. (b) L-I curves of all 4 channels at 50 °C.

chip.

The interposer measures approximately 5.8 mm x 2 mm and is based on POET's unique hybrid silicon photonics platform utilizing low loss and athermal dielectric waveguides. The waveguide width is narrow enough to maintain single transverse mode operation. Four waveguides are patterned on the interposer, each incorporating a lateral-coupled surface grating. The grating pitch defines the lasing wavelength while the frequency spacing between adjacent channels is designed to be 400 GHz. The targeted wavelengths are 1573.6 nm (CH1), 1576.8 nm (CH2), 1580.1 nm (CH3) and 1583.4 nm (CH4). The gratings are designed to provide high reflectivity (>90%) for optical feedback while maintaining appropriate reflection bandwidth for single longitudinal mode selection. A spot-size converter was integrated to optimize coupling between the gain chip and interposer waveguides. Integrated heater metal traces were fabricated above the grating region to allow fine wavelength tuning. Deep trenches were formed between adjacent waveguides to suppress thermal crosstalk. AuSn solder stubs are formed on the bonding region to provide both electrical connection and heat dissipation channels to the gain chips. Rectangular pedestals are patterned adjacent to the solder stubs to ensure accurate vertical alignment of the gain chips and interposer waveguides during flip-chip bonding. The gain-chip array is p-side down onto the optical interposer die. Each gain chip is 950 μ m long and the array contains four waveguides with a lateral pitch of 250 μ m, matching the interposer waveguide pitch. Anti-reflectivity (AR) coatings are applied to both facets. Ultra-low AR coating with reflectivity <0.1% was applied to the intra-cavity gain-chip facet to minimize reflection and was matched to refractive index of $n=1.53$ for subsequent potting with transparent epoxy. Partial-reflection coating was applied to the gain-chip output facet with reflectivity $\approx 10\%$ and matched to air refractive index. Z-stop structures are positioned on both sides of each waveguide. They are mesa structure fabricated

with precise vertical height control so that the optical mode from the gain chips aligns with the interposer waveguide when the z stops contact the interposer pedestals.

The gain-chip wafer was grown on 3-inch InP substrate by metal organic vapor phase epitaxy (MOVPE). The first epitaxial growth was terminated with InGaAsP quantum well layers and then a ridge waveguide was formed. Subsequent regrowth formed the p-type layers and also the semi-insulation InP layers to create a buried heterostructure for current confinement. The z-stop structures were later defined by wet etching. The interposer wafer was fabricated using 8-inch CMOS process flow. After fabrication, the wafer was back-grinded and stealth-diced into individual dies. Flip-chip bonding of the gain-chip array to the interposer was performed using an Amicra Nano die bonder, with the gain-chip front facets aligned to the interposer waveguide inputs. A transparent epoxy was applied at the coupling interface and UV-cured to achieve index matching. Electrical connection between the gain-chip backside metallization and the interposer cathode pads was subsequently established.

Results and Discussion

The hybrid integrated lasers were mounted on a thermoelectric cooler and characterized in continuous-wave (CW) mode at temperatures of 25 °C and 50 °C. For the light-current-voltage (L-I-V) measurement, current was injected into the gain chips using electrical probes, while the output power from the gain-chip output facet was measured using an integrating sphere. Fig. 2(a) shows the L-I-V characteristics of CH2 (targeted wavelength: 1576.8 nm) at 25 °C and 50 °C. The L-I-V curves exhibit no significant difference between these two temperatures. The threshold current is approximately 30 mA at 25 °C and 50 mA at 50 °C. Due to self-heating effect in the gain chip, the longitudinal modes shift toward longer wavelengths as the driving current increases. As a result, periodic mode hops appear in the L-I curves

due to detuning from the grating central wavelength. This behaviour is closely related to the kinks observed in the L-I curves, which correspond to transitions between longitudinal modes. In practical constant-power operation, the gain-chip current may be adjusted to compensate for performance variations, and the grating heater can be tuned accordingly to maintain alignment between the cavity mode and the Bragg grating, ensuring stable single-mode emission over a finite current tuning range. The single-pass coupling loss at the gain-chip/interposer interface is estimated to be approximately 1.7 dB. The maximum output power reaches 154 mW (21.9 dBm) at 500 mA and 25 °C, and 118 mW (20.7 dBm) at 475 mA and 50 °C. Since no thermal rollover is observed in the L-I curves, further power scaling, e.g. to 200 mW @25 °C and 150 mW @50 °C, is expected by increasing the drive current and improving the thermal management. Fig. 2(b) shows the L-I curves for all four channels at 50 °C. Except for CH2 which delivers slightly higher power, the remaining three channels exhibit similar performance.

For optical spectrum measurement, a lensed fiber was used to couple the light output from each channel into a Yokogawa AQ6370D optical spectrum analyzer with 0.02 nm resolution. Fig. 3(a) shows the superimposed spectrum of all the channels measured at 400 mA and 50 °C. The corresponding output powers measured using the integrating sphere are indicated alongside the spectrum. All channels operate in single-longitudinal-mode with side mode suppression ratios exceeding 50 dB. The frequency spacing between adjacent channels is controlled within $400 \text{ GHz} \pm 15 \text{ GHz}$ without active wavelength tuning. Fig. 3(b) shows the peak wavelength shift of CH4 as a function of heater power applied to the grating. The spectrum remains in single-mode throughout the tuning range and the peak wavelength exhibits an approximately linear dependence on heater power. With 100 mW heater power, the peak wavelength shifts by approximately 90 pm ($\sim 11 \text{ GHz}$). Therefore by utilizing the grating heaters, the channel spacing can be precisely controlled to 400 GHz.

The demonstrated output power represents one of the highest values reported for hybrid integrated lasers. Although this work is demonstrated in the L-band with a channel spacing of 400 GHz, the hybrid integration platform offers significant flexibility in wavelength selection and channel scalability. By engineering the grating design, which is decoupled from the gain section, operation can be extended to other wavelength bands such as the C-band and O-band, as well as to narrower channel spacings, e.g. 100 GHz and 50 GHz which have been demonstrated on POET's

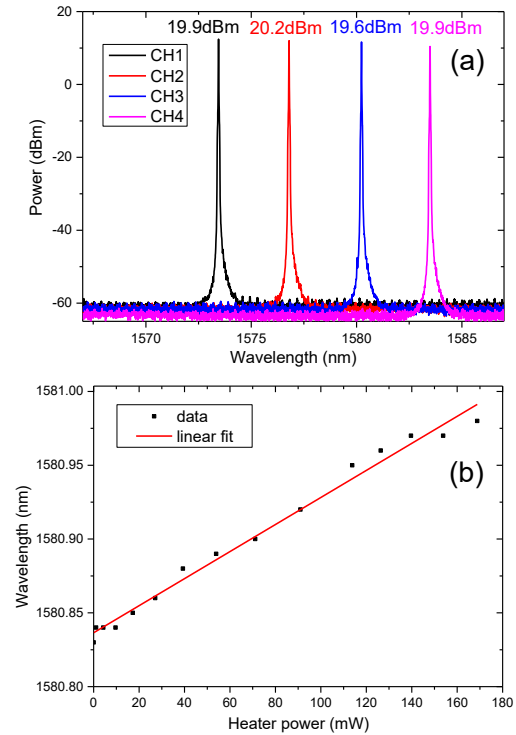


Fig. 3: (a) Optical spectrum of 4 channels, each driven at the current of 400 mA and 50 °C. (b) Peak wavelength shift as a function of grating heater power for CH4.

interposer platform and are desirable for DWDM applications. Another important application for silicon photonics is on-chip light sources. Hybrid integrated lasers offer flexibility in output configuration by redirecting the output from the gain-chip back facet to the interposer waveguide port. The optical power can be efficiently delivered through the silicon photonic circuit, potentially enabling on-chip high-power light sources for co-packaged optics applications.

Conclusion

A four-channel hybrid integrated laser array with 400 GHz wavelength spacing is demonstrated. More than 150 mW output power (@ 25 °C) is achieved from these L-band hybrid integrated lasers, representing a significant advancement in high-power silicon photonic light sources. This work paves the way for next-generation silicon photonic systems requiring high optical power, high transmission capacity, and scalable manufacturing.

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